

#ASLI ENGINEERING

Persistence

SWIPE

BY

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Persistence

Although Redis is an in-memory database.
it also provides periodic persistence

Two persistence formats
1. RDB
2. AOF

RDB persistence

RDB is a point-in-time snapshot of the Dataset

- used for snapshots
- single file output
- highly compact

How? Redis creates a fork and this new process dumps and creates the RDB file

No impact on Redis' performance
↓
given enough CPU exists

RDB creation should not be frequent operation



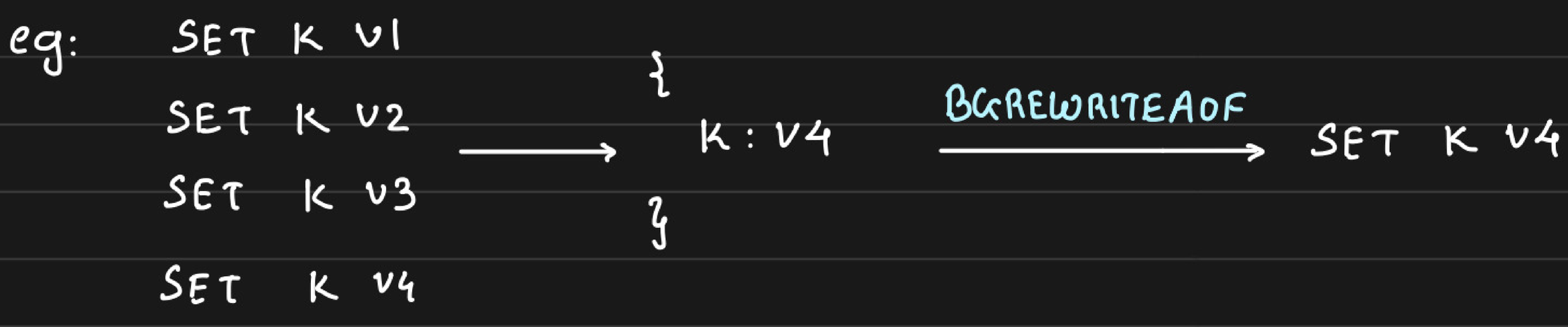
AOF persistence

AOF logs every single write operation to a file and replays it upon bootup to recreate the dataset.

The commands are logged is RESP format itself making is super simple to read and replay.

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Note: Periodically the entire AOF file is re-written for compaction



- AOF are much more durable
  - ↳ written every second (configurable)
  - Max data loss = 1 sec

APPEND ONLY FILE
↳ write perf.

redis-check-aof to check if AOF is valid (not corrupted)

How REWRITE happens in the background ?

Background thread rewrites to a temp AOF followed by a rename



Note: AOF files are bigger than RDB

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